

On the saturability of p -adic Lie groups

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Basic setup

Recall

The ring of **p -adic integers** $\mathbb{Z}_p$ is defined by

$$\mathbb{Z}_p := \varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \mathbb{Z}[[x]]/(x - p),$$

which is a discrete valuation ring, and the **p -adic numbers** $\mathbb{Q}_p$ are

$$\mathbb{Q}_p = \text{Frac}(\mathbb{Z}_p).$$

There is a natural topological ring structure on $\mathbb{Q}_p$ such that $p^n\mathbb{Z}_p$ for $n \in \mathbb{Z}$ forms an open neighborhood basis of zero.

Basic setup

Let $F/\mathbb{Q}_p$ be a finite field extension. We then have

- ▶ $\mathcal{O}_F \subseteq F$ the ring of integers,
- ▶ $\varpi_F \in \mathcal{O}_F$ a uniformizer,
- ▶ $\mathfrak{m}_F = \varpi_F \mathcal{O}_F \subseteq \mathcal{O}_F$ the maximal ideal,
- ▶ $k_F = \mathcal{O}_F/\mathfrak{m}_F$ the residue field,
- ▶ $e = e(F/\mathbb{Q}_p)$ the ramification index of F , and
- ▶ a valuation $v : F \rightarrow \frac{1}{e}\mathbb{Z}$ normalized so that $v(p) = 1$.

Basic setup

We want to study p -adic Lie groups G . These can be defined in the usual way using charts, atlases, etc.

Note

G is a compact p -adic Lie group if and only if G is isomorphic to a closed subgroup of $\mathrm{GL}_n(\mathbb{Z}_p)$ for some n .

Smoothness

Let V be a representation of a p -adic Lie group G , $K \leq G$ any subgroup. Then we write

$$V^K := \{v \in V \mid kv = v \text{ for all } k \in K\}.$$

Definition

A representation V of G is **smooth** if

$$V = \bigcup_{K \leq G} V^K,$$

where K runs over compact open subgroups of G . Let $\text{Mod}_{\mathbb{C}}(G)$, $\text{Mod}_{\mathbb{F}_p}(G)$ denote the categories of smooth $\mathbb{C}$ - and $\mathbb{F}_p$ -representations of G .

Exactness

Fact

Suppose that we are considering smooth *complex* representations of G , and let $K \leq G$ be a compact open subgroup. Then the functor

$$(-)^K : \text{Mod}_{\mathbb{C}}(G) \rightarrow \text{Vec}(\mathbb{C}), \quad V \mapsto V^K$$

is exact; that is, if $V \twoheadrightarrow W$ is G -equivariant, then $V^K \twoheadrightarrow W^K$ is still surjective.

This provides a useful tool for studying the complex representation theory of p -adic Lie groups.

Exactness

Warning!

This fails in characteristic p ! The functor

$$(-)^K : \text{Mod}_{\mathbb{F}_p}(G) \rightarrow \text{Vec}(\mathbb{F}_p), \quad V \mapsto V^K$$

is no longer exact.

In order to modify the techniques from the complex case, we must consider the right derived functors of $(-)^K$. These are written $H^i(K, -)$ for $i \geq 0$, and are also known as the **continuous group cohomology** of the compact p -adic Lie group K .

Motivation

Main goal

Compute the ring

$$H^*(G, \mathbb{F}_p) = \bigoplus_{i \geq 0} H^i(G, \mathbb{F}_p)$$

for G a compact p -adic Lie group.

Note

This is a (non-commutative) ring under the cup product.

History

- ▶ When G is a uniform pro- p group, i.e., when G is finitely generated and torsion-free, and $G/\overline{G^p}$ is abelian, we know

$$H^*(G, \mathbb{F}_p) \cong \bigwedge H^1(G, \mathbb{F}_p)$$

by [Laz65, Ch. V, Proposition 2.5.7.1]. (A good reference on uniform pro- p groups is [DDSMS99].)

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- ▶ If $H^*(G, \mathbb{F}_p)$ is finitely generated, then [Hen98, Theorem 1.4] shows that the canonical map

$$H^*(G, \mathbb{F}_p) \rightarrow \varprojlim_E H^*(Z_G(E), \mathbb{F}_p)$$

has finite kernel and cokernel, where E runs over nontrivial elementary abelian p -subgroups. He computes the right side explicitly for $G = \mathrm{GL}_n(\mathbb{Z}_p)$.

History

Key tool

When (G, ω) is a p -valued pro- p group, it has an associated $\mathbb{F}_p$ -Lie algebra $\mathfrak{g}$ with a multiplicative spectral sequence

$$E_1^{r,s} = H^{r,s}(\mathfrak{g}, \mathbb{F}_p) \implies H^{r+s}(G, \mathbb{F}_p) \quad (1)$$

by [Sor21, Theorem 5.5]. Here, $H^{r,s}(\mathfrak{g}, \mathbb{F}_p)$ denotes the graded $\mathbb{F}_p$ -Lie algebra cohomology of $\mathfrak{g}$, arising from the right derived functors of

$$(-)^{\mathfrak{g}} : \text{Mod}_{\mathbb{F}_p}(\mathfrak{g}) \rightarrow \text{Ab},$$

$$V \mapsto V^{\mathfrak{g}} := \{v \in V \mid Xv = 0 \text{ for all } X \in \mathfrak{g}\}.$$

The groups $H^{r,s}(\mathfrak{g}, \mathbb{F}_p)$ are easily computable!

Key example

In [Laz65, Ch. III, Exercice 3.2.6], it is shown that if $p - 1 > en$, then

$$I_1 = \{g \in \mathrm{GL}_n(\mathcal{O}_F) \mid g \equiv \begin{pmatrix} 1 & * & * \\ 0 & \ddots & * \\ 0 & 0 & 1 \end{pmatrix} \pmod{\mathfrak{m}_F}\}$$

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Recent application

Assume $e = 1$ and $p - 1 > n$, and recall $k = \mathcal{O}_F/\mathfrak{m}_F$. In a recent preprint [DH25], the authors use spectral sequence (1) applied to I_1 to compute $H^*(I, k_F)$, where

$$I = \{g \in \mathrm{GL}_n(\mathcal{O}_F) \mid g \equiv \begin{pmatrix} * & * & * \\ 0 & \ddots & * \\ 0 & 0 & * \end{pmatrix} \pmod{\mathfrak{m}_F}\},$$

and they prove that restriction is an isomorphism

$$H^*(\mathrm{GL}_n(\mathcal{O}_F), k_F) \xrightarrow{\sim} H^*(I, k_F).$$

Background: p -valuations

A good reference for this material is [Sch11].

Definition

Let G be a pro- p group. A **p -valuation** ω on G is a function

$$\omega : G \setminus \{1\} \rightarrow (0, \infty)$$

which, with the convention $\omega(1) = \infty$, satisfies the following for all $g, h \in G$:

1. $\omega(g) > \frac{1}{p-1}$;
2. $\omega(g^{-1}h) \geq \min\{\omega(g), \omega(h)\}$;
3. $\omega([g, h]) \geq \omega(g) + \omega(h)$;
4. $\omega(g^p) = \omega(g) + 1$.

For $r > 0$, write

$$G_r := \{g \in G \mid \omega(g) \geq r\}, \quad G_{r+} := \bigcup_{s>r} G_s = \{g \in G \mid \omega(g) > r\}.$$

Background: p -valuations

Associated to (G, ω) is an $\mathbb{F}_p[P]$ -Lie algebra

$$\mathrm{gr} G := \bigoplus_{r \geq 0} G_r / G_{r+1}$$

as well as an $\mathbb{F}_p$ -Lie algebra

$$\mathfrak{g} := \mathrm{gr} G \otimes_{\mathbb{F}_p[P]} \mathbb{F}_p = \mathrm{gr} G / P(\mathrm{gr} G).$$

Our pro- p group G will always have topology defined by ω (so the G_r form an open neighborhood basis of 1), and will always be of finite rank (so $\mathfrak{g}$ is finite dimensional).

Saturated p -valuations

Definition

We say a p -valuation ω is **saturated** if for $g \in G$,

$$g = h^p \iff \omega(g) > \frac{p}{p-1}.$$

Given a saturated group (G, ω) , one can define a $\mathbb{Z}_p$ -Lie algebra $L_\omega(G)_{\frac{1}{p-1}+}$, which comes with natural logarithm and exponent maps

$$G \begin{array}{c} \xrightarrow{\log} \\ \xleftarrow{\exp} \end{array} L_\omega(G)_{\frac{1}{p-1}+}.$$

Examples of p -valuations

Let $F/\mathbb{Q}_p$ be a finite extension, with valuation v normalized so $v(p) = 1$, and set $e = e(F/\mathbb{Q}_p)$.

Example 1

Fix any $a \in \mathbb{R}$. Then there is a saturated p -valuation ω on the additive group

$$G = \left\{ x \in F \mid v(x) > a + \frac{1}{p-1} \right\}$$

given by $\omega(x) := v(x) - a$.

In particular, if $p - 1 > e$, $\omega(x) := v(x)$ is itself a saturated p -valuation on $\mathfrak{m}_F$.

Examples of p -valuations

Example 2

Let $r \in \mathbb{Z}$ be the least integer strictly greater than $\frac{e}{p-1}$. Then there is a saturated p -valuation ω on the group of r -units

$$U_F^{(r)} = 1 + \mathfrak{m}_F^r$$

given by $\omega(1 + \varpi_F^r x) := v(\varpi_F^r x) = \frac{r}{e} + v(x)$. In particular, if $p - 1 > e$, then $r = 1$, so we get a saturated p -valuation on $U_F^{(1)}$.

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Note

In this case, the logarithm and exponent maps coming from the saturated pro- p group machinery recover the usual maps

$$U_F^{(r)} \begin{array}{c} \xrightarrow{\log} \\ \xleftarrow{\exp} \end{array} \mathfrak{m}_F^r.$$

Setup

Assumptions & Setup

1. Let $p \neq 2$ be an odd prime.
2. Let $F/\mathbb{Q}_p$ be a finite extension.
3. Let G be a connected reductive group defined over F (e.g., $G = \mathrm{GL}_n$; however G need not be split).
4. Assume that we have a maximally F^{ur} -split maximal torus $T \leq G_{F^{\mathrm{ur}}}$ which is weakly induced. That is, assume there's a basis of $X^*(T)$ on which the wild inertia group $\mathrm{Gal}(\overline{F}/F^t)$ acts via a permutation.

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Note

Our assumption on T holds automatically if G

- ▶ splits over a tame extension of F ,
- ▶ is simply connected, or
- ▶ is adjoint.

Background: Bruhat-Tits building

A good reference for this material is [KP23].

Key properties

- ▶ There exists a natural polysimplicial complex $\mathcal{B} = \mathcal{B}(G)$ called the **Bruhat-Tits building**.
- ▶ This is a union of **apartments** $\mathcal{A}(S)$ for $S \leq G$ maximal split tori, which are affine spaces over the real vector space $V(S_{\text{ad}}) = \mathbb{R} \otimes_{\mathbb{Z}} X_*(S_{\text{ad}})$ for $S_{\text{ad}} \leq G_{\text{ad}}$ the image of S in the adjoint group $G_{\text{ad}} = G/Z(G)$ (e.g., $G_{\text{ad}} = \text{PGL}_n$ if $G = \text{GL}_n$).
- ▶ The points of $\mathcal{A}(S) \subseteq \mathcal{B}$ are **valuations of the root datum** of (G, S) , which in particular are families of functions $(\phi_\alpha : U_\alpha(F) \rightarrow \mathbb{R} \cup \{\infty\})_{\alpha \in \Phi(G, S)}$.

Example building

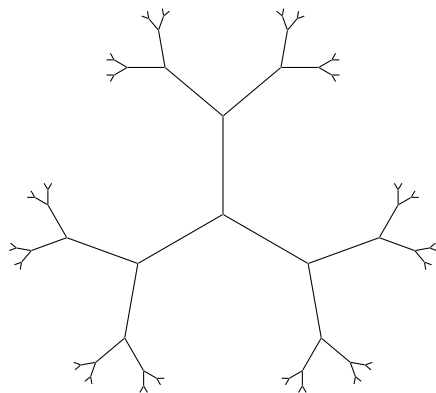


Figure: The Bruhat-Tits building of SL_2 (defined over $\mathbb{Q}_2$).

Background: Moy-Prasad filtrations

Associated to each $x \in \mathcal{B}(G)$ is a filtration of open compact subgroups

$$\{G(F)_{x,r}\}_{r \geq 0}.$$

These are the **Moy-Prasad filtration subgroups**. For $r \geq 0$, write

$$G(F)_{x,r+} := \bigcup_{s > r} G(F)_{x,s}.$$

Fact

If $r > 0$, then $G(F)_{x,r}$ is a pro- p group.

Background: Moy-Prasad filtrations

Let $G = \mathrm{GL}_n/F$. Write $e = e(F/\mathbb{Q}_p)$.

Example 1

If $x \in \mathcal{B}$ is the standard vertex, then for any $r \in \mathbb{Z}^+$,

$$G(F)_{x,r/e} = \{g \in \mathrm{GL}_n(\mathcal{O}_F) \mid g \equiv 1 \pmod{\mathfrak{m}_F^r}\}$$

is the r th principal congruence subgroup.

Example 2

If $x \in \mathcal{B}$ is in the dominant base alcove, then

$$G(F)_{x,0+} = \{g \in \mathrm{GL}_n(\mathcal{O}_F) \mid g \equiv \begin{pmatrix} 1 & * & * \\ 0 & \ddots & * \\ 0 & 0 & 1 \end{pmatrix} \pmod{\mathfrak{m}_F}\}$$

is the standard pro- p Iwahori, which is a pro- p Sylow of $\mathrm{GL}_n(\mathcal{O}_F)$.

Main theorem

Recall $p \neq 2$ and our torus $T \leq G_{F^{\text{ur}}}$ is weakly induced.

Theorem

For any $x \in \mathcal{B}(G)$,

$$\omega(g) := \sup\{r > 0 \mid g \in G(F)_{x,r}\}$$

defines a saturated p -valuation on $G(F)_{x, \frac{1}{p-1}+}$.

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This generalizes [LS24, Proposition 3.5], stated in [DH25] as:

Theorem

Assume G is split and $p-1 > eh$. Then for x the barycenter of the dominant base alcove, ω defines a saturated p -valuation on

$$G(F)_{x,0+} = G(F)_{x, \frac{1}{p-1}+}.$$

Here, h is the Coxeter number of G ; when $G = \text{GL}_n$, $h = n$.

Consequences

Say $p \neq 2$, and assume T is weakly induced. Group-theoretic consequences include:

Corollary

1. The pro- p group $G_{x, \frac{1}{p-1}+}$ is torsion-free.
2. For any $r > \frac{1}{p-1}$, $G_{x, r+1} = \{g^p \mid g \in G_{x, r}\}$.
3. For any $r \geq 1$, $G_{x, r}$ is a uniform pro- p group.

We also now have access to the spectral sequence

$$E_1^{r,s} = H^{r,s}(\mathfrak{g}, \mathbb{F}_p) \implies H^{r+s}(G(F)_{x, \frac{1}{p-1}+}, \mathbb{F}_p)$$

for computing the cohomology of $G(F)_{x, \frac{1}{p-1}+}$.

Future directions

- Compute the $\mathbb{Z}_p$ -Lie algebra L associated to $(G(F)_{x, \frac{1}{p-1}+}, \omega)$ by Lazard theory. There is a natural $\mathcal{O}_F$ -Lie algebra $\mathfrak{g}_{x, \frac{1}{p-1}+}$ endowed with a filtration of the same rank over $\mathbb{Z}_p$ as L , which initial computations suggest can be identified with L .

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- ▶ Compute the $\mathbb{F}_p[P]$ -Lie algebra $\mathrm{gr} G(F)_{x, \frac{1}{p-1}+}$ associated to $(G(F)_{x, \frac{1}{p-1}+}, \omega)$ by Lazard theory. Initial computations also suggest that the Moy-Prasad isomorphism (of abelian groups)

$$\bigoplus_{r > \frac{1}{p-1}} G(F)_{x,r} / G(F)_{x,r+} \xrightarrow{\sim} \bigoplus_{r > \frac{1}{p-1}} \mathfrak{g}(F)_{x,r} / \mathfrak{g}(F)_{x,r+}$$

is in fact an isomorphism of $\mathbb{F}_p[P]$ -Lie algebras.

Future directions

► If G is

1. split, or
2. quasisplit and semisimple,

and $p > c(G)$, then there is an open subset U of the dominant base alcove such that, for $x \in U$,

$$G(F)_{x,0+} = G(F)_{x,\frac{1}{p-1}+}.$$

Create a similar explicit bound for more general groups.

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Create a similar explicit bound for more general groups.

► Generalize the computation of [DH25] of the cohomology ring

$$H^*(I, k_F)$$

for I an Iwahori subgroup to more general settings; their argument required G to be split, F to be unramified, and $p - 1 > h$, in part due to relying on [LS24, Proposition 3.5].

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